SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2– CASE STUDY ANALYSIS

Course Code:	CSA65	**Course Title:**	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL2, BL3)	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	10% of CIA	**Total Marks:**	2 * 50 = 100 (2 Questions)
CO2 Statement:	*Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems (BL2, BL3)*		
Student Name:	ALLAPAREDDY JAHNAVI	**Reg. No.:**	192472214
Section / Batch:	CSE AI /2024	**Date:**	

Instructions to Students

- Answer **2 questions**. Each question carries **50 marks**. **Total: 100 Marks.**
- Demonstrate a **deep understanding of the concepts** relevant to the given case study.
- Identify and analyze the **key problems, requirements, challenges, and constraints** presented in the case.
- Apply appropriate **technical concepts, methods, and approaches** to propose and justify a suitable solution.
- Discuss the **real-world implications, applications, benefits, limitations, and challenges** of the proposed solution.
- Provide **thorough, evidence-based, and well-structured analysis**, supported by suitable examples, diagrams, architectures, or workflows wherever applicable.
- Present the analysis with a **clear and professional structure**, using appropriate headings and logical organization.
- Use **relevant references and proper citation practices** and maintain clear academic and technical writing.

Code of Conduct

I ALLAPAREDDY JAHNAVI (192472214) that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

SECTION A – CASE STUDY ANALYSIS

1. Enterprise Customer Support Assistant

A large e-commerce company wants to develop an AI assistant to answer customer queries about orders, refunds, products, and delivery. The company plans to use a pre-trained LLM.

Analyze the case and propose an appropriate LLM-based architecture. Explain the role of the transformer architecture, prompting, function calling, and structured output in the proposed system.

2. University Academic Assistant

A university wants to build an AI assistant that answers student questions about courses, examinations, attendance, and academic regulations.

Analyze the requirements and recommend whether prompt engineering, fine-tuning, transfer learning, or a combination of these approaches should be used. Justify your decision.

3. Domain-Specific Legal Assistant

A legal organization wants to adapt a general-purpose LLM to understand legal terminology and generate responses in a consistent legal format.

Analyze whether full fine-tuning, transfer learning, few-shot prompting, or PEFT/LoRA would be most appropriate. Discuss the advantages and limitations of your selected approach.

4. Medical Report Analysis System

A hospital wants an AI system that summarizes medical reports into predefined sections such as findings, observations, and recommendations.

Analyze how structured prompting can be used to produce consistent outputs. Discuss the risks of hallucination and suggest appropriate prompt constraints and evaluation methods.

5. Financial Advisory Assistant

A financial company wants an AI assistant that answers questions about financial products and generates explanations for customers.

Analyze how a pre-trained LLM could be adapted for this domain. Discuss the roles of pre-training, fine-tuning, prompt engineering, and structured prompting.

6. Customer Query Classification

An organization receives thousands of customer queries that must be classified into Billing, Technical Support, Account, and General categories.

Analyze whether zero-shot, one-shot, or few-shot prompting would be most suitable. Design an approach and justify your choice.

7. Inconsistent AI Responses

A company notices that its LLM gives different answers to similar customer queries.

Analyze the possible causes of inconsistency and propose prompt engineering techniques involving role definition, context, constraints, examples, and output formatting to improve consistency.

8. Few-Shot Product Classification

An e-commerce company wants to classify products into categories where the product descriptions are highly varied.

Analyze the suitability of few-shot prompting compared with zero-shot prompting. Explain how examples should be selected and how the resulting prompts can be evaluated.

9. One-Shot Email Classification

A company wants to classify incoming emails as Urgent or Non-Urgent but has very limited labeled examples.

Analyze how one-shot prompting could be applied. Explain the advantages, limitations, and expected impact on classification accuracy.

10. Complex Reasoning Assistant

A financial application needs an LLM to solve multi-step reasoning problems.

Analyze how a reasoning-oriented prompting strategy could be used. Discuss how the prompt can encourage reliable reasoning while ensuring that the final answer follows a predefined format.

11. Prompt Engineering for Text Summarization

A news organization wants to summarize long articles into exactly five bullet points.

Design and analyze an appropriate prompt. Explain the role of context, instructions, constraints, and expected output format.

12. Multilingual Translation Assistant

A global company wants an AI assistant to translate customer messages between multiple languages while preserving technical terminology.

Analyze the prompt requirements and recommend an appropriate prompting strategy. Explain how examples and output constraints can improve translation quality.

13. Code Generation Assistant

A software company wants an LLM to generate Python programs from natural-language requirements.

Analyze the challenges associated with code generation and propose an appropriate prompt design. Discuss how structured prompting and output constraints can improve the generated code.

14. Resume Screening System

A recruitment company wants to use an LLM to analyze resumes and identify suitable candidates.

Analyze how structured prompting could be used to extract candidate name, skills, experience, education, and suitability score. Discuss potential limitations and evaluation criteria.

15. Contract Clause Extraction

A legal company wants to extract termination clauses, payment clauses, and confidentiality clauses from contracts.

Analyze how prompt engineering and structured output can be combined to build this application. Discuss how the system should handle missing clauses.

16. AI Interview Assistant

A recruitment organization wants an LLM to generate interview questions based on a candidate's skills and experience.

Analyze how few-shot prompting and structured prompting could be used. Explain how the prompt can ensure that questions remain relevant and unbiased.

17. Enterprise Knowledge Assistant

An enterprise wants an AI assistant that uses its internal documents but does not want to train a new LLM from scratch.

Analyze the advantages of using a pre-trained LLM with transfer learning or prompt engineering. Recommend the most suitable approach and justify it.

18. Limited Computational Resources

A startup wants to customize a large LLM for customer support but has limited GPU resources and budget.

Analyze why full fine-tuning may be unsuitable. Explain how LoRA and PEFT can address the problem and recommend an appropriate adaptation strategy.

19. LoRA-Based Healthcare Model

A healthcare organization wants to adapt an existing LLM to generate standardized medical summaries.

Analyze how LoRA can be used for the task. Compare it with full fine-tuning in terms of trainable parameters, computational requirements, storage, and deployment.

20. PEFT Strategy Selection

A company wants to customize a large model using a small domain-specific dataset.

Analyze whether full fine-tuning or PEFT would be more appropriate. Consider computational resources, dataset size, model size, and expected performance.

21. Pre-Training vs Fine-Tuning

A company plans to create a specialized LLM for customer service.

Analyze what knowledge the model would obtain during pre-training and what knowledge or behavior could be adapted through fine-tuning. Explain why training from scratch may not be necessary.

22. Transfer Learning for Education

An educational technology company wants to adapt an existing LLM for generating educational explanations and quizzes.

Analyze how transfer learning can be applied. Explain the relationship between the pre-trained model, domain-specific data, and the adapted model.

23. LLM Architecture Selection

A company is deciding whether to use a transformer-based LLM for its conversational AI application.

Analyze the role of tokenization, embeddings, attention mechanisms, transformer layers, and output generation in producing the final response. Explain why transformer architecture is suitable for the application.

24. Function Calling for Banking

A banking assistant should answer account queries and perform actions such as checking balance and initiating transactions.

Analyze how function calling can be incorporated into the system. Identify suitable functions, parameters, validation requirements, and security considerations.

25. Function Calling for E-Commerce

An online shopping assistant should search products, check availability, and place orders.

Analyze how function calling can allow the LLM to interact with external systems. Design an appropriate workflow and discuss how incorrect or ambiguous requests should be handled.

26. Function Calling for University Services

A university assistant should check attendance, retrieve examination information, and generate student notifications.

Analyze which functions should be provided to the LLM and how the system should determine when a function needs to be called. Discuss suitable structured outputs.

27. Sequential Function Calling

A travel assistant receives the request: “Find hotels in Chennai for two nights and calculate the estimated cost.”

Analyze how multiple function calls can be performed sequentially. Explain the workflow from the user's query to the final response.

28. Secure Function Calling

A banking AI assistant is allowed to access customer accounts and initiate fund transfers.

Analyze the security risks associated with function calling and propose safeguards such as parameter validation, authentication, confirmation, authorization, and human approval.

29. Choosing an LLM API

A startup wants to develop an AI chatbot and is considering OpenAI, Gemini, and Hugging Face.

Analyze the three options based on model availability, API access, customization, deployment, integration, cost considerations, and function/structured-output capabilities. Recommend a suitable option for the startup.

30. LLM API Migration

An organization has built an application using one LLM API but wants to migrate to another provider.

Analyze the technical and application-level challenges involved in migration. Discuss prompt compatibility, model behavior, API differences, structured outputs, error handling, and testing.

31. Hugging Face Deployment

A research team wants greater control over model selection and customization and is considering Hugging Face models.

Analyze the advantages and challenges of using Hugging Face compared with hosted LLM APIs. Recommend a suitable deployment approach based on the scenario.

32. Prompt Evaluation Framework

An organization has created five different prompts for the same customer-support task but does not know which one performs best.

Design an evaluation approach using criteria such as accuracy, relevance, consistency, completeness, format adherence, and response quality. Explain how the best prompt should be selected.

33. Prompt Refinement

An AI assistant frequently provides unnecessarily long answers and sometimes ignores important instructions.

Analyze the weaknesses of the existing prompt and propose a refined prompt using clear instructions, context, constraints, and output formatting. Explain how each modification improves performance.

34. Hallucination in LLM Applications

A company discovers that its AI assistant sometimes provides information that is not supported by the supplied context.

Analyze the possible causes of hallucination and recommend prompt engineering and system-level strategies to reduce it. Explain how the effectiveness of these strategies can be evaluated.

35. Structured Output for Enterprise Reports

An enterprise wants an LLM to generate reports containing fixed fields such as summary, key findings, risk level, recommendations, and priority.

Analyze how structured prompting can ensure consistent output. Discuss what should happen when information required for a field is unavailable.

36. Comparing Prompting Strategies

A company wants to use an LLM for three different tasks: sentiment classification, document classification, and complex problem solving.

Analyze which prompting strategy—zero-shot, one-shot, few-shot, or reasoning-oriented prompting—would be appropriate for each task. Justify your choices.

37. Prompt Injection Scenario

An enterprise chatbot receives user-provided documents and instructions. Some documents contain instructions that attempt to change the chatbot's behavior.

Analyze the potential prompt-injection risk and propose prompt design and system-level safeguards to reduce the risk.

38. Complete AI Customer Service System

An enterprise wants to build a customer-service system using a pre-trained LLM, prompt engineering, function calling, structured output, and an LLM API.

Analyze and design an end-to-end architecture. Explain how the LLM, prompts, functions, external systems, API, and structured responses interact. Discuss edge cases and evaluation methods.

39. Enterprise LLM Customization Strategy

A large organization wants to customize an LLM for HR, finance, and technical-support applications. It has a large amount of domain data but limited computational resources.

Analyze whether prompting, transfer learning, fine-tuning, or PEFT/LoRA should be used. Propose a suitable strategy for each domain and justify your decisions.

40. Integrated Generative AI Application

A company wants to develop an enterprise AI assistant capable of understanding natural-language queries, generating responses, calling external tools, producing structured outputs, and adapting to a specialized business domain.

Analyze the complete solution covering**:**

- LLM architecture
- Pre-training and fine-tuning
- Transfer learning
- Prompt engineering
- Zero-shot/one-shot/few-shot prompting
- Reasoning-oriented prompting
- Structured prompting
- Function calling
- LoRA/PEFT
- LLM API selection
- Prompt evaluation
- Security and edge-case handling

Recommend an end-to-end architecture and justify the technologies, techniques, and design decisions selected for the proposed system**.**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Depth	30%	Deep, accurate understanding of pre-training/fine-tuning/RLHF concepts	Good understanding with minor gaps	Superficial understanding of concepts	Misunderstands core concepts
Real-World Implications	25%	Draws well-reasoned real-world implications and comparisons	Reasonable implications drawn	Weak or generic implications	No meaningful analysis presented
Analytical Rigor	20%	Analysis is thorough, evidence-based, and well-structured	Mostly thorough analysis	Partial analysis with gaps in evidence	Superficial, unstructured analysis
Report Structure	15%	Report/slides professionally structured, easy to follow	Clear structure with minor issues	Structure unclear in places	Poorly structured submission
Citation & Academic Writing	10%	Properly cites sources; clear academic writing	Mostly cited, minor lapses	Inconsistent citation practice	No citation or referencing

Marks Evaluation:

Question No.	Conceptual Depth(30%)	Real-World Implications (25%)	Analytical Rigor (20%)	Report Structure (15%)	Citation & Academic Writing(10%)	Total Marks (100)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

QUESTION 1 – ENTERPRISE CUSTOMER SUPPORT ASSISTANT

Case Study: A large e-commerce company wants to develop an AI assistant to answer customer queries about orders, refunds, products, and delivery. The company plans to use a pre-trained LLM.

Question: Analyze the case and propose an appropriate LLM-based architecture. Explain the role of the transformer architecture, prompting, function calling, and structured output in the proposed system.

1. Introduction

An Enterprise Customer Support Assistant is an AI-based system designed to automatically answer customer questions and help customers complete shopping-related activities. E-commerce companies receive many queries related to order tracking, product information, payment status, refunds, cancellations, and delivery.

Traditional customer support depends heavily on human agents and predefined responses. These systems may take more time to respond and may not understand complex customer questions. A pre-trained Large Language Model (LLM), integrated with prompt engineering, function calling, and structured output, can support a more flexible customer service system.

2. Problem Identification and Analysis

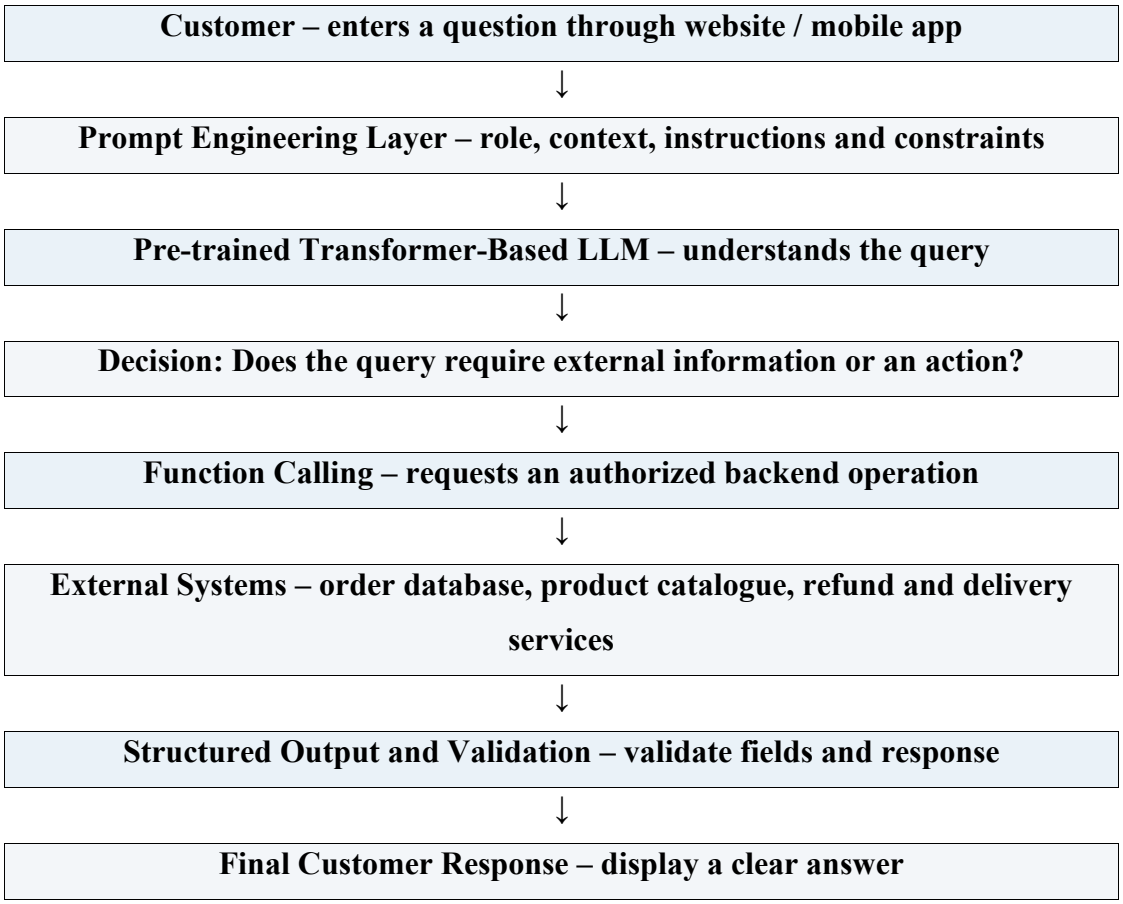
- High query volume: Thousands of customers may contact support every day.
- Delayed responses: Human agents may take time to answer customer questions.
- Order tracking difficulties: Customers need updated information about their orders.
- Refund management: Customers require accurate information about refund status.
- Product-related queries: Customers ask about prices, specifications, and availability.
- Inconsistent responses: Different support agents may provide different answers.
- Limited availability: Traditional customer support may not operate 24 hours a day.
- Proposed solution: Develop an LLM-based customer support assistant connected to the company's order management, product catalogue, payment, and delivery systems.

3. Proposed LLM-Based Architecture

The proposed architecture combines a customer interface, prompt engineering, a pre-trained transformer-based LLM, function calling, backend services, structured output validation, and a response interface.

Component	Function
Customer Interface	Accepts customer questions through a website or mobile application.
Prompt Engineering Layer	Adds the assistant role, context, instructions, and constraints.
LLM API	Sends the prompt to a pre-trained transformer-based language model.
Function Calling Layer	Allows the LLM to request information or actions from external business systems.
Backend Services	Connects to order management, product catalogue, refund, and delivery systems.
Structured Output Layer	Formats responses into predefined fields such as JSON.
Response Interface	Displays the final answer to the customer.

4. Architecture Diagram



For general questions, the LLM may answer directly using its general language knowledge. For order-

specific questions, it should request verified information through an authorized backend function.

5. Role of Transformer Architecture

A transformer is the main architecture used by many modern LLMs. It helps the model process relationships between tokens and generate contextually relevant responses.

Component	Function in Customer Support
Tokenization	Converts customer messages into smaller units called tokens.
Embeddings	Converts tokens into numerical representations.
Positional information	Helps the model represent the order of tokens.
Self-attention	Identifies relationships between different words in a query.
Transformer layers	Process contextual information through multiple layers.
Output generation	Produces a natural-language response.

Example: For the query "Where is my order and when will it arrive?", the model identifies the request as an order-tracking question. It can then request order details from the backend.

6. Role of Prompt Engineering

Prompt engineering is the process of designing clear instructions to guide an LLM to produce accurate and relevant responses.

You are an e-commerce customer support assistant.

Your responsibilities are:

1. Answer customer questions politely.
2. Help customers track their orders.
3. Provide product information.
4. Explain refund and cancellation procedures.
5. Use authorized tools to retrieve order information.
6. Never invent order status, prices, or delivery dates.
7. If information is unavailable, inform the customer.
8. Ask for clarification when order details are missing.

Keep responses simple, clear, and professional.

Prompt engineering improves the system through role definition, context, instructions, constraints, and output-format requirements.

7. Role of Function Calling

Function calling allows an LLM to interact with external applications and databases. The model identifies

when a tool is required and generates a structured request. The backend validates and executes the request, then returns the result to the LLM.

Function	Parameters	Purpose
track_order	Order ID	Retrieves current order status.
search_product	Product name	Searches the product catalogue.
check_stock	Product ID	Checks product availability.
check_refund	Order ID	Retrieves refund status.
cancel_order	Order ID and customer authorization	Requests order cancellation.

Example function request:

```
{
"function": "track_order",
"arguments": {
"order_id": "ORD1025"
}
}
```

Example backend result:

```
{
"order_id": "ORD1025",
"status": "Shipped",
"expected_delivery": "5 November 2026"
}
```

The assistant can then say: Your order has been shipped and is expected to arrive on 5 November 2026. The backend must authenticate the customer, authorize access, validate parameters, and execute the function safely. The LLM should not directly access the database.

8. Role of Structured Output

Structured output ensures that the LLM returns information in a predefined format, such as JSON. This makes responses easier for backend applications to validate and process.

```
{
"query_type": "order_tracking",
"order_id": "ORD1025",
"order_status": "Shipped",
"delivery_status": "In Transit",
"response": "Your order has been shipped and is on
```

Makes responses consistent.

Helps backend applications process information.

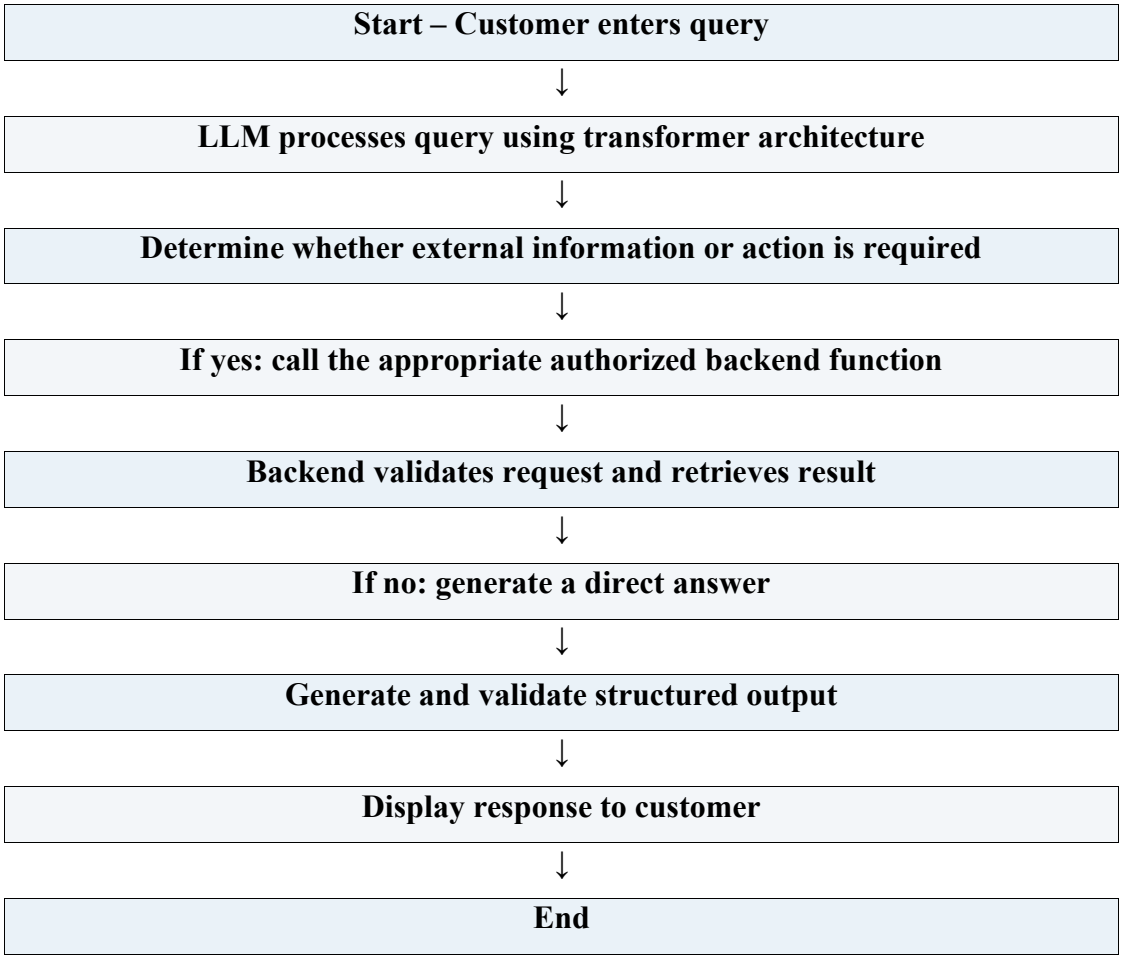
Reduces formatting errors.

Supports automated workflows.

Makes system integration easier.

The system should validate the output against a defined schema and handle missing or invalid fields.

9. Complete Workflow



10. Advantages and Limitations

Advantages	**Limitations**
Provides 24/7 customer support.	API usage may increase operational costs.
Reduces repetitive work for human agents.	LLMs may generate incorrect information.
Provides faster responses.	Requires secure database integration.
Handles multiple types of customer queries.	Complex queries may need human assistance.
Provides consistent response formats.	Depends on backend system availability.

11. Security and Risk Management

- Authenticate customers before showing personal order information.
- Use role-based authorization for backend functions.
- Validate all function parameters.
- Ask for explicit confirmation before cancellation or refund actions.
- Protect personal and payment information.
- Maintain logs for auditing and error investigation.
- Escalate sensitive or unresolved issues to human agents.

12. Conclusion

The proposed Enterprise Customer Support Assistant uses a pre-trained transformer-based LLM integrated with prompt engineering, function calling, and structured output. Transformer architecture enables natural-language understanding, while prompt engineering guides the model to respond professionally. Function calling connects the LLM with external e-commerce systems to retrieve accurate information and perform authorized operations. Structured output ensures consistent and machine-readable responses. This architecture can improve customer satisfaction and reduce support workload. Security, data accuracy, validation, and human supervision remain essential.

QUESTION 2 – UNIVERSITY ACADEMIC ASSISTANT

Case Study: A university wants to build an AI assistant that answers student questions about courses, examinations, attendance, and academic regulations.

Question: Analyze the requirements and recommend whether prompt engineering, fine-tuning, transfer learning, or a combination of these approaches should be used. Justify your decision.

1. Introduction

A University Academic Assistant is an AI-based application that helps students obtain information about academic activities. Students regularly ask questions about course details, examination schedules, attendance requirements, assignments, results, and university regulations.

Traditional academic support depends on faculty members, administrative offices, and university websites. A pre-trained LLM can understand student questions and generate natural-language responses. However, a general-purpose LLM may not contain the latest university-specific information. The university therefore needs to combine suitable adaptation techniques with access to official academic information.

2. Problem Identification and Analysis

Repeated student queries: Students frequently ask similar questions about attendance, examinations, and

courses.

Information availability: Students may find it difficult to locate correct academic information.

Outdated information: General-purpose LLMs may not know current university regulations.

Different question formats: Students may ask the same question in different ways.

Administrative workload: Faculty and office staff spend time answering routine queries.

Personalized requirements: Some questions require information specific to a student's course, semester, or academic record.

Accuracy and reliability: Incorrect answers about examination rules or attendance may negatively affect students.

Proposed solution: Build a University Academic Assistant using a pre-trained LLM, prompt engineering, and retrieval from university documents. Fine-tuning can be added later if the university needs more consistent specialized response behavior.

3. Recommended Approach

Recommended strategy: a combination of a pre-trained LLM, prompt engineering, and Retrieval-Augmented Generation (RAG), with transfer learning or fine-tuning introduced only when justified by evaluation.

Pre-trained LLM: Provides general language understanding and response generation.

Prompt engineering: Guides the assistant to answer in a university-specific manner.

Transfer learning: Adapts an existing model to academic tasks when domain-specific training is needed.

Fine-tuning: Further improves task-specific behavior when sufficient high-quality examples are available.

RAG: Supplies current university regulations, course details, and examination information to the LLM.

The university should begin with prompt engineering and retrieval because they use existing academic documents without requiring the university to train a model from scratch. Fine-tuning should be considered only if testing shows that prompting and retrieval are insufficient.

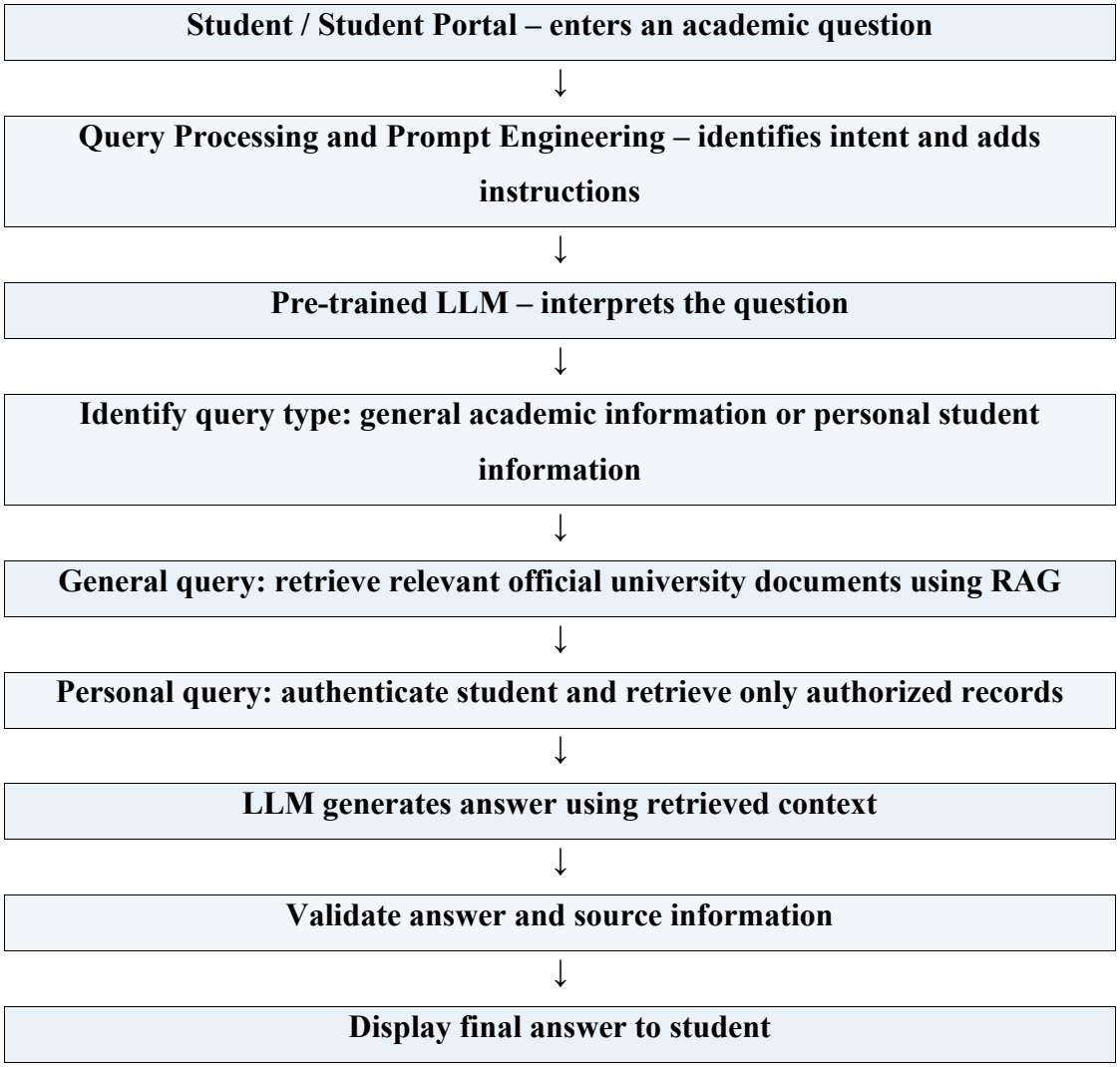
4. Proposed System Architecture

The architecture includes a student interface, query processing, prompt engineering, a pre-trained LLM, academic document retrieval, authorized student-data access, and response validation.

Component	**Description**
Student Interface	Allows students to enter questions through a website or application.
Query Processing	Identifies the topic and intent of the student's question.
Prompt Engineering	Provides the LLM with role, context, and answer

	instructions.
Pre-trained LLM	Understands questions and generates natural-language responses.
Academic Knowledge Base	Stores university rules, course details, examination information, and notices.
Retrieval System	Searches the knowledge base for relevant information.
Student Information System	Provides authorized student-specific information such as attendance.
Response Validation	Checks the answer for relevance, completeness, and correct formatting.

5. Architecture Diagram



6. Analysis of Different Adaptation Techniques

A. Prompt Engineering

Prompt engineering is the process of designing clear instructions and context for an LLM to generate the required response.

You are a University Academic Assistant.

Your responsibilities are:

1. Answer questions about courses and subjects.
2. Explain examination schedules and procedures.
3. Provide attendance requirements based on official rules.
4. Answer questions about academic regulations.
5. Use the supplied university documents as your source.
6. Do not invent dates, marks, rules, or attendance details.
7. If the information is unavailable, ask the student

to contact the academic office.

Give answers in simple and clear language.

Advantages: easy to implement, does not require model training, can be modified quickly, and helps maintain a consistent response style.

Limitations: does not automatically provide new university knowledge; ambiguous prompts may produce inconsistent results; and the model may still generate unsupported information.

B. Fine-Tuning

Fine-tuning means further training a pre-trained LLM using a smaller, task-specific dataset.

Student Question	**Expected Answer**
What is the attendance requirement?	Refer to the applicable official attendance regulation.
How can I register for examinations?	Follow the examination registration procedure published by the university.
Where can I find course details?	Check the official course information or academic portal.
How can I check my results?	Use the university's authorized result portal.

Advantages: can improve performance on specialized academic tasks and teach consistent response patterns. Limitations: requires suitable training data, computational resources, testing, and periodic review; it does not automatically keep university information up to date.

C. Transfer Learning

Transfer learning uses knowledge already learned by a pre-trained model and adapts it to a new task or domain. The university can start with an existing general-purpose LLM and adapt it to academic communication, course-related questions, and university terminology.

Advantages: reuses knowledge from an existing model, avoids training from scratch, and can improve domain-specific performance. Limitations: requires suitable examples and resources, and does not guarantee knowledge of current university rules.

D. Retrieval-Augmented Generation (RAG)

RAG retrieves relevant information from external documents and provides it as context to the LLM before answer generation. When a student asks about an examination regulation, the system searches official university documents and provides relevant content to the LLM.

Advantages: uses updated university documents, reduces dependence on the model's stored knowledge, can ground answers in official sources, and allows documents to be updated without retraining the LLM. Limitations: depends on document quality and retrieval accuracy, poorly organized documents may produce irrelevant results, and private student information requires access controls.

7. Comparison of Techniques

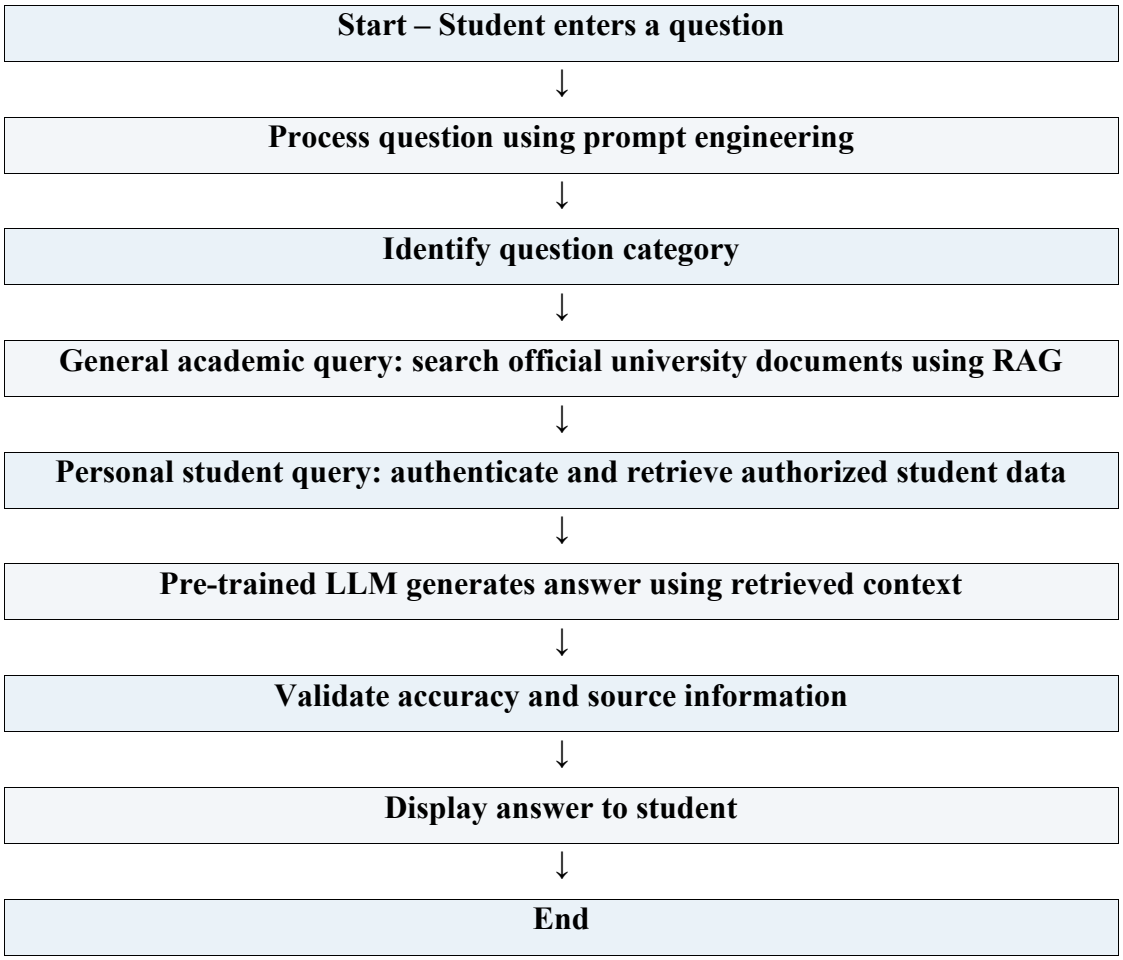
Criteria	**Prompt Engineering**	**Fine-Tuning**	**Transfer Learning**	**RAG**
Main purpose	Guide responses	Adapt task behavior	Adapt an existing model	Retrieve current information
Training required	No	Yes	Usually adaptation/training	No LLM training required
Cost	Low	Medium to high	Medium to high	Low to medium; depends on infrastructure
Updating university rules	Change prompt or sources	May require retraining	May require adaptation	Update documents
Domain specialization	Instructions	Training examples	Adapted model	Retrieved knowledge
Best use	Style and instructions	Repeated specialized tasks	Academic-domain adaptation	Current university information

8. Recommended Implementation Strategy

- Stage 1 – Select a pre-trained LLM with suitable language understanding and deployment options.

- Stage 2 – Prepare university documents such as course syllabi, examination rules, attendance regulations, academic calendars, and official notices.
- Stage 3 – Implement RAG by dividing documents into smaller sections, creating embeddings, and storing them in a searchable vector database.
- Stage 4 – Design prompts defining the assistant's role, answer requirements, source restrictions, and response format.
- Stage 5 – Add authorized student-data access through secure backend functions. Only authenticated students should access their own records.
- Stage 6 – Evaluate accuracy, relevance, consistency, and whether responses are supported by official documents.
- Stage 7 – Consider fine-tuning if the assistant still struggles with academic terminology or specialized response patterns.

9. Workflow / Flowchart



10. Advantages and Limitations of the Proposed Solution

Advantages	Limitations
Provides 24/7 academic assistance.	Requires regular document updates.
Reduces repeated administrative queries.	May retrieve incorrect or incomplete documents.
Provides quick access to academic information.	Requires secure access to student records.
Supports different types of student questions.	Complex cases may require faculty support.
Can be expanded to different departments.	Model and infrastructure costs may increase.
Uses official documents as answer context.	Human verification remains important for critical decisions.

11. Evaluation Methods

Evaluation Criteria	Purpose
Accuracy	Checks whether the answer is factually correct.
Relevance	Checks whether the response addresses the student's question.
Completeness	Checks whether important details are included.
Consistency	Checks whether similar questions receive similar answers.
Source correctness	Verifies whether the response is supported by official documents.
Response time	Measures how quickly the assistant answers.
Security	Checks whether private student information is protected.

Example test case: Input – 'What is the examination registration procedure?' Expected behavior – retrieve the current official examination instructions. Evaluation – compare the response with the official university notice and check that the assistant has not invented dates or requirements.

12. Conclusion

The University Academic Assistant can be developed using a pre-trained LLM combined with prompt engineering and Retrieval-Augmented Generation. Prompt engineering controls the assistant's behavior, while RAG provides current information from official university documents. Transfer learning and fine-tuning can be introduced when additional academic-domain adaptation is necessary. This combined approach balances accuracy, flexibility, cost, and maintainability. Security, official-source verification, and human supervision must be maintained for important academic decisions.